

POOJA SAHU

Poojads2@illinois.edu | (217) 637 3023 | www.linkedin.com/in/sahu-pooja/ | Champaign, IL | <https://www.poojasahu.tech/>

EDUCATION

University of Illinois Urbana-Champaign, School of Information Science **Champaign, IL**
Master of Science in Information Management (CGPA: 4.0/4.0) Aug 2025 – May 2027 (Expected)
Relevant Coursework: Advance AI Web-App Development, Database Design & Prototyping, Data Statistics

Narsee Monjee Institute of Management Studies (NMIMS) **Mumbai, India**
B.Tech. Electronics and Telecommunication + MBA in Technology Management (CGPA: 3.53/4.0) Jul 2018 – May 2023
Major: Business Intelligence & Analytics | Minor: Marketing Analytics

SKILLS

- **Programming & Backend:** Python, SQL, FastAPI, Django, Node.js, Javascript
- **Gen AI:** LLM, RAG, Agentic AI, Prompt Engineering, LLM Evaluation, Conversational AI, NLP, Kubernetes, MCP
- **AI Frameworks:** LangChain, LangGraph, LlamaIndex, NeMo Guardrails
- **Retrieval & Knowledge Systems:** Embeddings, Vector Search, FAISS, Semantic Similarity, Structured Info. Extraction
- **Machine Learning:** scikit-learn, XGBoost, Regression, Classification, Computer Vision, YOLOv5, Feature Engineering, Cross-Validation, Model Evaluation, Hypothesis Testing
- **Data & Cloud:** Pandas, NumPy, MySQL, Apache Spark (PySpark), Supabase, Google Cloud Platform, Cloudera
- **Tools:** Git, GitHub, Jupyter, Streamlit, Power BI, Tableau, Groq

PROFESSIONAL EXPERIENCE

COUNTRY Financial **Champaign, IL**
Data Science Intern May 2026 – Present

- Developed a **PolicyPro multi-layer LLM evaluation framework** using deterministic validation, semantic similarity, evidence-based assessment, and risk classification to systematically identify response-quality issues and reduce manual evaluation effort for enterprise AI-assistant outputs
- Built an automated AI evaluation framework combining **deterministic validation, semantic similarity, evidence-based assessment, and risk classification** to test LLM response quality and reliability
- Built **AI-powered document validation system** using OCR, PDF layout analysis, and template comparison, automating key insurance billing quality-assurance checks and accelerating the identification of document discrepancies.
- Developed memory-efficient preprocessing and feature-engineering pipelines for **14.5M+ insurance records**, supporting large-scale actuarial modeling and automated data-quality validation

ACCENTURE **Mumbai, India**
Security Delivery Analyst Jun 2023 – Jul 2025

- Developed and configured **enterprise security and risk-management** workflows in ServiceNow IRM/SecOps, translating technical requirements into automated controls, validations, notifications, and application workflows that reduced manual review effort by 20% (~200 hours/month)
- Built Python-based data **validation and anomaly-detection** workflows across **50K+ enterprise records**, identifying processing failures and improving data accuracy and SLA adherence
- Engineered integrations, scheduled data feeds, workflow automation, and testing across enterprise risk systems including ServiceNow and RSA Archer, supporting secure and reliable production processes
- Collaborated across 3 engineering squads through requirements analysis, testing, defect resolution, release planning, and iterative software delivery in an **Agile environment**

CAPGEMINI **Mumbai, India**
Data Science Intern May 2021 – June 2021

- Developed a real-time weapon detection model using YOLOv5, training and fine-tuning the **object detection pipeline** on 3,000+ custom-labeled images, achieving **98.48% classification accuracy** to support security surveillance use cases
- Automated data-validation and reconciliation workflows in Python, reducing recurring reporting errors by 15–20%

PROJECTS

sust-AI-naible (Multi-Agent Carbon-Aware Scheduler) | Python, Multi-Agent Systems, Generative AI

- Designed a multi-agent AI architecture coordinating specialized Grid Forecast, Scheduler, Workload, and Policy agents to support carbon-aware compute scheduling.
- Developed orchestration logic for agents to exchange contextual information and collaboratively generate workload scheduling recommendations

SRB-RAG-Powered WhatsApp AI Assistant | Python, FastAPI, FAISS, Twilio, Embeddings

- Built a retrieval-augmented generation (RAG) chatbot integrating FastAPI, FAISS vector search, and WhatsApp messaging through Twilio to retrieve contextual information and generate grounded responses.
- Designed the backend retrieval and conversational workflow to connect user queries with relevant indexed knowledge.
- Designed deterministic comparison pipelines and machine-readable validation outputs to identify inconsistencies across multi-page documents and communication workflows without relying on hard-coded templates

Large-Scale Insurance Data Processing & Risk Modeling | Python, Machine Learning, Data Pipelines

- Developed memory-efficient preprocessing and feature-engineering pipelines for 14.5M+ records and 180+ variables, using chunk-based processing to support large datasets under local compute constraints.
- Automated data-quality checks, special-value handling, feature filtering, variable selection, and downstream modeling workflows to create a reusable pipeline for large-scale insurance analytics.